Taiwan Artificial Intelligence Industry Association (TAIA)

January 26, 2026

Management & Technology Summer Institute (M&TSI) Admissions Committee
Jerome Fisher Program in Management & Technology
University of Pennsylvania

Re: Recommendation for Aaryan Samanta (M&TSI Summer 2026)

Dear Members of the Admissions Committee,

I am writing in my capacity as Chairperson of the Taiwan Artificial Intelligence Industry Association (TAIA) to offer my strongest recommendation for Aaryan Samanta's application to the University of Pennsylvania's Management & Technology Summer Institute (M&TSI) for Summer 2026. This letter is provided exclusively for his M&TSI application.

I had the opportunity to observe Aaryan directly during the Taiwan Roadshow (August 10–13, 2025), where he served as Founder and President of NextGenAI and presented on AI in Medical Imaging and AI Ethics, with a particular emphasis on accountability and responsible deployment. The Roadshow involved high-stakes cross-cultural engagements under real operational constraints, including typhoon-related disruptions, frequent transitions between audiences, and limited preparation windows. Aaryan consistently performed at an exceptional level across these conditions.

In particular, at a formal session hosted by Southern Taiwan University of Science and Technology (STUST) on August 12, 2025 (1:30–3:00 PM), Aaryan delivered a structured, college-level presentation to a high-level audience that included senior university leadership (including the president and vice presidents), R&D leadership, faculty members, and parents. In my role, I evaluated not only the technical content, but also his ability to communicate with senior stakeholders—clarity of thought, judgment under pressure, and the maturity to navigate interdisciplinary questions. Aaryan distinguished himself on all three.

Aaryan's most notable strength is his ability to bridge deep technical thinking with management and venture-oriented reasoning. He explained complex AI capabilities in medical imaging with accuracy and discipline, and then translated those capabilities

into decision-relevant considerations: adoption constraints, workflow integration, accountability boundaries, risk tradeoffs, and stakeholder incentives. In other words, he did not treat AI as a purely technical artifact—he framed it as a system that must be responsibly designed, governed, and deployed. This is precisely the mindset that M&TSI aims to develop through its integrated technology–management curriculum and its team-based prototyping and go-to-market process.

Beyond his intellectual strength, Aaryan shows the qualities of a high-impact contributor in team innovation settings. He takes ownership naturally, anticipates what a group needs to move forward, and elevates collaboration by bringing structure to ambiguous problems. During discussions with mixed audiences—professors seeking rigor, administrators focused on institutional implications, and parents and students looking for relevance—he adjusted his level of abstraction without diluting substance. This adaptability reflects both communication intelligence and authentic leadership.

Aaryan also demonstrated a rare level of professionalism and composure in live Q&A. He responded in a methodical, evidence-based manner: clarifying assumptions before answering, addressing ethical concerns directly rather than deflecting, and maintaining a respectful tone even under challenging questioning. These are not rehearsed "presentation skills"; they are indicators of mature judgment and integrity—qualities that matter in entrepreneurial environments where decisions have real consequences.

Based on my direct observation and interactions, I am confident Aaryan will thrive in M&TSI's rigorous college-level coursework and intensive team prototyping and pitching experience. He will contribute technical competence, disciplined thinking, and a strong ethical compass—while also bringing the initiative, collaboration, and cross-cultural maturity that strengthen any cohort. I recommend him to you without reservation.

Please feel free to contact me if I can provide additional context or specific examples.

Sincerely,

Dr. Li Chieh Lin
Chairperson, Taiwan Artificial Intelligence Industry Association (TAIA)

Letter of Recommendation

Applicant: Aaryan Samanta
Recommender: Dr. Tian Yishuai, Postdoctoral Fellow, Tsinghua University **Email:** mohan0620@163.com
Date: December 18, 2025

To the Admissions Committee,

It is my great pleasure to recommend Aaryan Samanta for admission to your undergraduate program. As a postdoctoral fellow at Tsinghua University working heavily in AI, optimization theory and computational research, I have mentored undergraduates, graduate students and postdoctoral fellows at Stanford and elsewhere. In that context, Aaryan is one of the most research-ready intellectually mature high schoolers I have ever met (top 1%).

It was clear from our first conversation that Aaryan had an unusual mastery of higher level technical material. He doesn't simply "use" machine learning tools - he knows the math they are built on, the limits of algorithms, and the tradeoffs in optimization. His work mixing quantum inspired optimization with graph neural network architectures is the kind of conceptual blend I sometimes see in strong undergrad researchers, but rarely.

One instance that stands out happened during a pair debugging session on a GNN-genetic algorithm hybrid for tumor prediction. Aaryan caught on while tweaking the loss mid-way through, pointed out how the loss was optimizing for accuracy on a majority demographics while completely ruining performance on more diverse cohorts & at 15 drew out and proposed a fairness-weighted surrogate loss that resolved this but then triggered 2 hours of talking on equity-weighted surrogates which shifted our focus to equity in omics data.

This gave rise to a paper from Aaryan as lead author at a peer-reviewed international AI conference (AIAM 2025). Rather than replicating an existing paper format, he independently argued for the merit of a hybrid genetic algorithm-GNN ensemble, justified his design choices theoretically, implemented the full pipeline of tasks, and wrote up and analyzed the results with academic rigor. This isn't a throwaway peer published paper; this is research with insight and independence. Fewer than 1% of U.S. high school students ever publish in a peer-reviewed AI venue, and even fewer do as lead author (Aaryan is top 1% for turning shiny tech into virtue).

Besides research Aaryan is extremely good at algorithmics by all means. Getting dual perfects in the USACO 2025 US Open (Gold and Silver divisions) means being in the top 0.1% of US competitive programmers who must be fast but also have great algorithmic reasoning ability and resilience. And perfect AMC 10/12s, AIME qualifs seal the deal.

But what really makes Aaryan stand out isn't his raw talent. It's his agency. He's always asking himself the right questions: why is this model failing?What assumptions are baked in?

How could this generalize better? Can we make this more efficient? These are the questions of a researcher, not a student doing homework exercises - the kind of kid who'll one day end up in an MIT CSAIL ethical AI lab or a Stanford HAI societal frontiers lab. When I work with Aaryan, I feel more like an elder statesman colleague than a mentor.

Equally noteworthy is his awareness of responsibility and leadership - as founder of the NextGenAI International Organization & AI Ethos non profit at the mere age of 15, and now, representative for over 75+ global members, he has fought for ethical AI education in underserved communities and their rural U.S inner cities, Taiwan and Thailand reaching 75+ members in 3 countries, and using Assembly AI's bias-detection curricula, he bridges the data gaps that leave behind 350 million with rare diseases. Such awareness is critical of our modern day science and engineering - more than any other time in history, technology boasts the power to save society from a myriad of ills - without the compass of ethical boundaries, we will not go far.

In sum, Aaryan is among the strongest (top 1%) pre-college researchers I have mentored, and I have no doubt he will excel at a rigorous, research-oriented place like MIT or Stanford or peer school. I recommend him highly, and without reservation. He is ready to contribute meaningfully and reshape undergrad research from day one. I'd be happy to talk further if that would be helpful.

Sincerely,

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Dr. Tian Yishuai
Postdoctoral Fellow, Tsinghua University